FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Main)-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. Use relation $\Delta V = -\int \vec{E}.d\vec{r}$

2. A

Sol. Heat produced = work done by battery – energy change in capacitor

3. D

Sol. Initial current is $\frac{\varepsilon}{R}$ and final steady current is $\frac{\varepsilon}{2R}$.

4. C

Sol. Basic concept: Kirchoff's law

5. B

Sol. Using $P = \frac{\rho RT}{M}$

6. D

Sol. Maximum temperature is at midpoint of straight line.

7. C

Sol. Area under PV cycle gives work done.

8. B

Sol. $k_{eff} = \frac{2k}{3}$

$m_{eff} = \frac{m}{2}$

9. B

Sol. Speed $v = \omega\sqrt{A^2 - x^2}$ and $|\vec{a}| = \omega^2 |\vec{x}|$

10. C

Sol. Using $T = \frac{YA}{\ell}.\Delta\ell$, $f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$

11. B

Sol. Basic concept of Doppler's effect.

12. D

Sol. Using $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

13. C

Sol. Resultant $\vec{v}$ should be along $\overrightarrow{AC}$.

14. B

Sol. Using constraint: $a_A = 2a_B$ and writing equation of motion & solving we get the answer.

15. C

Sol. There is no friction between blocks until 8 kg block moves.

16. C

Sol. Basic concept.

17. C

Sol. After 1st polarizer intensity becomes $\frac{I_0}{2}$ and after that changes by $\cos^2\theta$ times.

18. B

Sol. Using Moseley's aw.

19. C

Sol. Basic concept: $h\nu - \phi = KE$

20. B

Sol. Basic concept.

21. A

Sol. $P_{total} = \left(1 + \frac{m^2}{2}\right)P_{carrier}$

22. C

Sol. $\vec{L} = m\vec{r}_{cm} \times \vec{v}_{cm} + I_{cm}\vec{\omega}$

23. D

Sol. Total energy equals sum of self energy of two shells and their mutual energy.

24. C

Sol. Current in inductor lags voltage by $\frac{\pi}{2}$.

25. A

Sol. Using $\frac{q}{C} = vB\ell$ and $F - i\ell B = m\frac{dv}{dt}$

26. A

Sol. Magnetic field inside cavity is uniform, so particle goes in helical path.

27. B

Sol. Basic concept of potentiometer.

28. D

Sol. $\frac{1}{2}I_A\left(\frac{u}{L}\right)^2 = 2mg\left(\frac{3L}{2}\right)$

29. A

Sol. $X = A^{\frac{1}{4}}.Y^{\frac{1}{16}}$

$$\frac{\delta X}{X} = \frac{1}{4}\frac{\delta A}{A} + \frac{1}{16}\frac{\delta Y}{Y}$$

30. C

Sol. Torque about CM = $I_{cm}.\alpha$

Chemistry

PART – II

SECTION – A

31. A

Sol. $n-l=4$

$l=3$

i.e $n=7$

$\therefore 7f$

32. A

Sol. $M+X_2 \rightarrow MX_2;\ 3MX_2+X_2 \rightarrow M_3X_8$

$M_3X_8+4N_2CO_3 \rightarrow 8NX+4CO_2+M_2O_4$

$\frac{206}{103}\times\frac{1}{8}\times 3\times 56=42$ gm

33. B

Sol. When S_8 decomposes to S_2; size $\downarrow$;b $\downarrow$

34. B

Sol. $R(r)=kr^l\ Le^{-r/na_o}$

$\therefore l=1$

$n=2$

$2P_z$ [for P_z, xy plane is nodal plane; $\cos 90^o=0$]

35. B

Sol. $X(s) \rightleftharpoons A(g)+2B(g)$

$P_1 \quad 2P_1+2P_2$

$Y(s) \rightleftharpoons 2B(g) + C(g)$

$2P_1+2P_2 \quad P_2$

Total pressure = $P_1+P_2+2P_1+2P_2=3(P_1+P_2)$

$k_{p_1}=4P_1(P_1+P_2)^2;k_{p_2}=4P_2(P_1+P_2)^2$

$k_{p_1}+k_{p_2}=4(P_1+P_2)^3;P_1+P_2=1.5\times 10^{-1}$

36. B

Sol. $C(s)+O_2(g) \rightarrow CO_2(g)$

$715+500-1750-\text{R. E.}=-390$

R.E $=-145$ kJ/mol

37. B

Sol. Q 1 mol of $KHC_2O_4.H_2C_2O_4$ will have basicity = 3

$\therefore$ moles of $Ba(OH)_2$ required = 3/2

38. A

Sol. $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$T_2 = \dfrac{300}{8^{0.33}} = 150K$

$W = \dfrac{nR\Delta T}{\gamma - 1} = \dfrac{-1\times 2\times 150}{0.33} = -900 \text{ cal}$

39. D

Sol. since O^{-2} ions occupy ccp; $\therefore O^{-2} = 4$

Number of $A = 8\times\dfrac{1}{6} = 4/3$

Number of $B = 4\times\dfrac{1}{3} = 4/3$

$\therefore$ Formula $A_{\frac{4}{3}}B_{\frac{4}{3}}O_4 = ABO_3$

40. A

Sol. After 15 min. concentration of B = B/2 and that of A = B/2.

41. A

Sol. $E_a\left(\text{over all}\right) = \dfrac{2}{3}\left(Ea_1 + Ea_2 - Ea_3\right)$

$= \dfrac{2}{3}(180 + 80 - 50)$

$= 140 \text{ kJ/mol}$

42. A

Sol. $^{27}_{13}Al + ^{4}_{2}He \xrightarrow{\text{(i)}} ^{30}_{14}Si + ^{1}_{1}H(X)$

$^{27}_{13}Al + ^{4}_{2}He \xrightarrow{\text{(ii)}} ^{30}_{15}P + ^{1}_{0}n(Y)$

$\downarrow$

$^{30}_{14}Si + ^{0}_{+1}e(Z)$

43. C

Sol. If it will not dissociate then will not show abnormal value.

44. B

Sol. For pure ethanol $X_{\text{methanol}} = 0$

$\therefore P = 135$

45. B

Sol. Fact.

46. B

Sol. From Si to Cl in 3rd period, decreasing size increases strength of $p\pi - d\pi$ bond with oxygen.

47. A

Sol. ICl is polar and has stronger I.M.F.

48. A
Sol. Fact.

49. A
Sol. As charge on C.A. $\downarrow$, $C-O$ bond strength $\uparrow$, or $C-O$ bond order $\uparrow$, or C – O bond length $\downarrow$.

50. C
Sol. Fact.

51. B
Sol. Does not contain any element of symmetry.

52. B
Sol. Epimers are the compounds having different orientation of groups about a particular carbon.

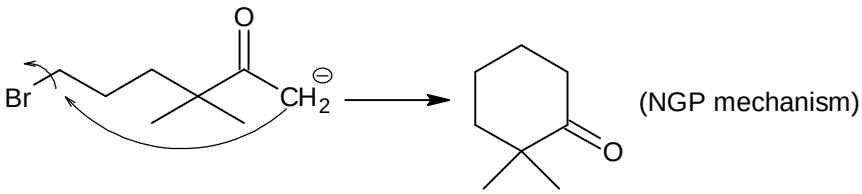
53. A
Sol. In cyclic compounds only the groups existing in same plane can undergo elimination.

54. A
Sol. Hydroboration – oxidation reaction.

55. C
Sol. LAH does not reduce C=C.

56. B
Sol. NGP mechanism.

57. B
Sol.

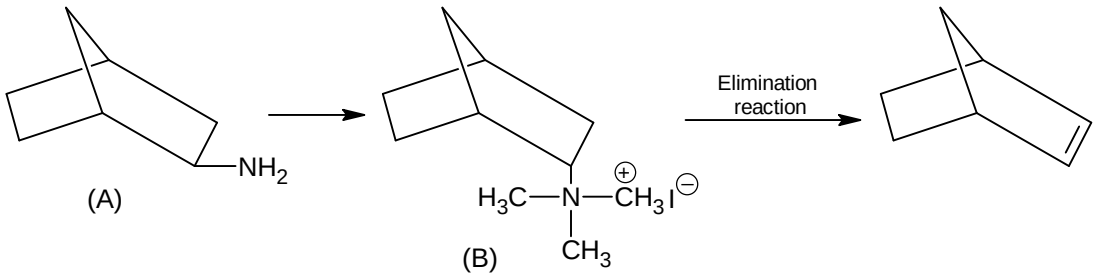


(NGP mechanism)

58. D
Sol. β-keto carboxylic acid undergo elimination most easily.

59. C
Sol. Fact.

60. B
Sol.



Mathematics

PART – III

SECTION – A

61. C

Sol. Let hyperbola is $(x-2y)(x+y+1)-\lambda=0$

since it touches x-axis $(y=0)$

$$x(x+1)-\lambda=0$$

$D=0 \Rightarrow \lambda=-\frac{1}{4}$ so $\alpha=4, \beta=1$

62. C

Sol. $f(y)=e^{-y^2}\int_0^1 e^x\left(x^3-2xy+3x^2-2y\right)dx$

$=e^{-y^2}\left[e^x\left(x^3-2xy\right)\right]_0^1=e^{-y^2}\left(e(1-2y)-0\right)=e^{(1-y^2)}(1-2y)$

$\therefore \int_0^1 e^y f(y)dy=\int_0^1 e^{y+1-y^2}(1-2y)dy$

$=\int_1^1 e^t dt=0$.

63. D

Sol. $f(x)=x^2-2x+2\sin\alpha$

$f(-1)\,f(1)<0$

$(2\sin\alpha-1)(3+2\sin\alpha)<0 \Rightarrow \sin\alpha<\frac{1}{2}<\frac{5\pi}{6}<\alpha<2\pi$

So length is $2\pi-\frac{5\pi}{6}=\frac{7\pi}{6}$.

64. A

Sol. Range is $\left\{0, \sqrt{\pi}, \sqrt{\frac{\pi}{2}}\right\}$.

65. A

Sol. $K={}^{10}C_5$ and exponent of 2 in ${}^{10}C_5$ is 2.

66. D

Sol. Let mid point is (α, β). So equation of AB is $T=S_1 \Rightarrow \frac{x\beta+y\alpha}{2}-2=\alpha\beta-2$

$x=\frac{2\alpha\beta-y\alpha}{\beta}$.

$y^2-4\left(2\alpha-y\frac{\alpha}{\beta}+1\right)=0$

$y^2+y\frac{4\alpha}{\beta}-4(2\alpha+1)=0$

$D=0 \Rightarrow 16\frac{\alpha^2}{\beta^2}+16(2\alpha+1)=0$

$\frac{x^2}{y^2}+(2x+1)=0$

67. D

Sol. Let $z = x + iy$

$\sqrt{x^2 + (y+1)^2} - x - i(y-1) = 0$

$\Rightarrow y = -1$ and $x = \lambda, \lambda \geq 0$

$z = \lambda - i$

68. D

Sol. A_1 : – white ball is drawn from bag 1

A_2 : – Red ball is drawn from bag 1

B : – 2 balls are drawn from bag 2 and found to be red.

$P(B) = P(A_1).P(B/A_1) + P(A_2)P(B/A_2)$

$$\Rightarrow P(A_1/B) = \frac{\frac{3}{5}.\frac{{}^5C_2}{{}^{10}C_2}}{\frac{3}{5}.\frac{{}^5C_2}{{}^{10}C_2} + \frac{2}{5}.\frac{{}^6C_2}{{}^{10}C_2}} = \frac{1}{2}$$

69. C

Sol. $$a_n = \frac{2n+3}{4.(1+2+..n+1)^2} = \frac{2n+3}{\frac{4(n+1)^2\ (n+2)^2}{4}}$$

$$= \frac{(n+2)^2 - (n+1)^2}{(n+1)^2\ (n+2)^2} = \frac{1}{(n+2)^2} - \frac{1}{(n+1)^2}$$

$$S_n = -\left(\frac{1}{(n+2)^2} - \frac{1}{2^2}\right) \Rightarrow S_\infty = \frac{1}{4}.$$

70. C

Sol. $$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = f'(0) + 2x$$

$\Rightarrow f(x) = x^2 \Rightarrow I_1 = I_2 < I_3$

71. C

Sol. $$S = \lim_{n\to\infty} \frac{1}{n} \sum \frac{1}{\left(3\sqrt{\frac{r}{n} + 4}\right)^2 \sqrt{\frac{r}{n}}}$$

$$= \int_0^4 \frac{dx}{\sqrt{x}(3\sqrt{x} + 4)^2} = \frac{1}{10}$$

72. D

Sol. G is $\left(x + y\frac{dy}{dx}, 0\right)$

$OG = 2x \Rightarrow y\,dy = x\,dx \Rightarrow y^2 - x^2 = C$ which is rectangle hyperbola.

73. A

Sol. Required area $= 2\int_0^{\sqrt{3}} \left(2x^2 + 9 - 5x^2\right)dx = 12\sqrt{3}$ sq. unit .

74. D

Sol. $AM \geq GM \Rightarrow \dfrac{n\sin\alpha + \dfrac{1}{n}\cos\alpha}{2} \geq \sqrt{\sin\alpha\cos\alpha}$

75. D

Sol. $P(\bar{B}/\bar{A}) = \dfrac{P(\bar{B}\cap\bar{A})}{P(\bar{A})} = \dfrac{1-P(A\cup B)}{0.6} \Rightarrow P(A\cup B) = .64,\ P(B/A) = \dfrac{P(B\cap A)}{P(A)} \Rightarrow P(A\cap B) = 0.36$

$\Rightarrow P(B) = 0.6,\ P(A/B) = \dfrac{P(A\cap B)}{P(B)} = .6$

$P(A/\bar{B}) = \dfrac{P(A\cap\bar{B})}{P(\bar{B})} = 0.1$

76. D

Sol. Any point on $y = mx$ is (t, mt)

Equation of chord under middle point (t, mt)

$T = S_1 \Rightarrow m^2t^2 - 2(2m+1)t + 8 = 0$

For two chords,

$D > 0 \Rightarrow 4m^2 - 4m - 1 < 0$

77. A

Sol. $x_1 + x_2 + x_3 = 11$.....(1)

$x_1 = 0, 1, . . ,,$

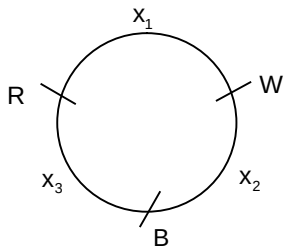
$x_2 = 0, 1 \ldots , ,$

$x_3 = 0, 1, \ldots , ,$

So require number of arrangements

$= \dfrac{3!}{3} \times$ Number of solution of (1)

$= 2! \times {}^{13}C_{11}$



78. C

Sol. Put x = 1, y = h in the relation

$2\left(\dfrac{f(1+h)-f(1)}{h}\right) - \dfrac{(f(1-h)-f(1))}{-h} + 3h = 2$

take limit $h \to 0$

$2f'(1) - f'(1) = 2 \qquad \Rightarrow f'(1) = 2$

79. C

Sol. There is no loss of generality in considering the order of bi's as $b_1 < b_2 < b_3 \cdots < b_n$ also given that $f(a_1) \leq f(a_2)\cdot f(a_{2n})$. Now suppose number of preimages of every b_i are x_i in numbers.

Therefore $x_1 + x_2 + \cdots + x_n = 2n$ where $1 \leq x_i \leq n+1 \rightarrow (1)$

Number of solutions of (1) is ${}^{2n-1}C_{n-1}$ or ${}^{2n-1}C_n$

80. C

Sol. $\angle B + \angle C = 150^o$

$\cos(B+C)=\frac{-\sqrt{3}}{2} \Rightarrow \cos B\cos C-\sin B\sin C=\frac{-\sqrt{3}}{2}$

Also $\frac{1}{2}ab\sin C=\frac{\sqrt{3}}{4}a^2 \Rightarrow \sin B\sin C=\frac{\sqrt{3}}{4}$

$\Rightarrow \tan B\tan C=-1$

$\Rightarrow B=90^o+C$

Therefore $\angle B=120^o,\ \angle C=30^o$

81. B

Sol. $\alpha+\beta=-b/a,\ \alpha\beta=\frac{c}{a}$

Determinants

$$D=\begin{vmatrix} 1 & \cos(\beta-\alpha) & \cos\alpha \\ \cos(\alpha-\beta) & 1 & \cos\beta \\ \cos\alpha & \cos\beta & 1 \end{vmatrix}=\begin{vmatrix} \cos\alpha & \sin\alpha & 0 \\ \cos\beta & \sin\beta & 0 \\ 1 & 0 & 0 \end{vmatrix}\begin{vmatrix} \cos\alpha & \sin\alpha & 0 \\ \cos\beta & \sin\beta & 0 \\ 1 & 0 & 0 \end{vmatrix}=0$$

hence will have non trivial solution.

82. A

Sol. $\frac{\sin A}{\sin C}+\frac{\cos A}{\cos C}=\frac{2}{\sqrt{3}}$ & $A+C=150^o$

$\Rightarrow C=30^\circ,\ A=120^\circ$

$a:b:c=\sin A:\sin B:\sin C=\sqrt{3}:1:1$

$K\left(2+\sqrt{3}\right)=4+2\sqrt{3} \Rightarrow K=2$

$\Delta=\frac{1}{2}\cdot 2\cdot 2\frac{\sqrt{3}}{2}=\sqrt{3}$ sq. unit

83. B

Sol. $f(x)=2(x-i)(x+i)(x-3+i)(x-3-i)$

$f(1)=2.(1+1)(4+1)=2-a+2b+c-d$

$\Rightarrow \quad 2b+c=18+a+d$

$f(0)=2(1)(i-3)(-i-3)$

$-d=20 \quad \Rightarrow \quad d=-20$

$f(-1)=2(-1-i)(-1+i)(-4+i)(-4-i) \quad =2+a+2b-c-d$

$\Rightarrow \ a+2b-c=46$ and $2b+c-a=-2$

$\Rightarrow$ b = 11

therefore, b + d = −9

84. C

Sol. $\arg\left(z(1+\bar{z})\right)+\arg\left(\frac{z\bar{z}}{z(1-\bar{z})}\right)=0$

$\Rightarrow \ \arg z+\arg(1+\bar{z})+\arg\bar{z}-\arg(1-\bar{z})=0$

$\Rightarrow \ \arg\left(\frac{1+\bar{z}}{1-\bar{z}}\right)=0 \quad \Rightarrow \ \frac{1+\bar{z}}{1-\bar{z}}=k>0 \quad \Rightarrow \ \bar{z}=\frac{k-1}{k+1} \quad \Rightarrow \ |\bar{z}|<1$

85. B

Sol. If $\left[\frac{1}{3}+\frac{n}{90}\right]=0 \quad \Rightarrow \quad 0 \le \frac{1}{3}+\frac{n}{90}<1$

$\Rightarrow \quad 0 \le \frac{(30+n)}{90}<1 \quad \Rightarrow \quad 0 \le (n+30)<90$

$\Rightarrow \quad 1 \le n<60$

and If $\left[\frac{1}{3}+\frac{n}{90}\right]=1 \quad \Rightarrow \quad 1 \le \frac{1}{3}+\frac{n}{90}<2$

$\Rightarrow \quad 90 \le (n+30)<180 \quad \Rightarrow \quad 60 \le n<150$

$\therefore \quad \sum_{n=1}^{59}\left[\frac{1}{3}+\frac{n}{90}\right]+\sum_{n=60}^{k}\left[\frac{1}{3}+\frac{n}{90}\right]=21$

$\Rightarrow \quad 0+\sum_{n=60}^{k}\left[\frac{1}{3}+\frac{n}{90}\right]=21 \quad \Rightarrow \quad \sum_{n=60}^{k}\left[\frac{1}{3}+\frac{n}{90}\right]=21$

which is possible only when k = 80. $\left[\because \quad \left[\frac{1}{3}+\frac{n}{90}\right]=1\right]$

86. B

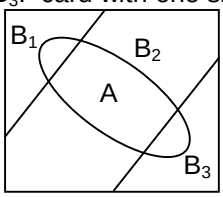
Sol. Let us consider the following events.

A: card shows up black

B_1: card with both sides black

B_2: card with both sides white

B_3: card with one side white and one black



$P(B_1)=\frac{2}{10},\ P(B_2)=\frac{3}{10},\ P(B_3)=\frac{5}{10}$

$P(A/B_1)=1,\ P(A/B_2)=0,\ P(A/B_3)=\frac{1}{2}$

$$P(B_1/A)=\frac{\frac{2}{10}\times 1}{\frac{2}{10}\times 1+\frac{3}{10}\times(0)+\frac{5}{10}\times\frac{1}{2}}=\frac{4}{4+5}=\frac{4}{9}$$

87. A

Sol. Let CD be the position of man at any time t. Let BD = x, then EC = x. Let $\angle ACE=\theta$

Given, AB = 41.6 m, CD = 1.6 m and $\frac{dx}{dt}=2$ m/s.

$AE = AB - EB = AB - CD = 41.6 - 1.6 = 40$ m

We have to find $\frac{d\theta}{dt}$ when x = 30 m

From ΔAEC, $\tan\theta=\frac{AE}{EC}=\frac{40}{x}$ (1)

Differentiating w.r.t. to t, $\sec^2\theta\frac{d\theta}{dt}=\frac{-40}{x^2}\frac{dx}{dt}$

$\Rightarrow \sec^2\theta\dfrac{d\theta}{dt}=\dfrac{-40}{x^2}\times 2$

$\Rightarrow \dfrac{d\theta}{dt}=\dfrac{-80}{x^2}\cos^2\theta=-\dfrac{80}{x^2}\dfrac{x^2}{x^2+40^2}\quad\left[\because \cos\theta=\dfrac{x}{\sqrt{x^2+40^2}}\right]$

$\Rightarrow \dfrac{d\theta}{dt}=-\dfrac{80}{x^2+40^2}$

when x = 30, $\dfrac{d\theta}{dt}=-\dfrac{80}{30^2+40^2}=-\dfrac{4}{125}$ radian/s.

88. B

Sol. $T_n=(n+1)(n\omega+1)(n\omega^2+1)$

$=(n+1)(n^2-n+1)$

$=n^3+1$

$\therefore\quad S_n=\sum T_n=\sum n^3+\sum 1$

$=\left(\dfrac{n(n+1)}{2}\right)^2+n$

89. D

Sol. We have $\dfrac{1}{\sin^2\theta\cos^2\theta}+\dfrac{2}{\sin^2\theta}=8,\ \sin\theta\neq 0,\ \cos\theta\neq 0$

$\Rightarrow\quad 1+2\cos^2\theta=8\sin^2\theta\cos^2\theta=8\cos^2\theta(1-\cos^2\theta)$

$\Rightarrow\quad 8\cos^4\theta-6\cos^2\theta+1=0$

$\Rightarrow\quad (4\cos^2\theta-1)(2\cos^2\theta-1)=0$

$\Rightarrow\quad \cos^2\theta=1/4=\cos^2(\pi/3)$ or $\cos^2\theta=1/2=\cos^2(\pi/4)$

$\Rightarrow\quad \theta=n\pi\pm(\pi/3)$ or $\theta=n\pi\pm(\pi/4),\ n\in Z$

Hence, for $0\le\theta\le\pi/2$, $\theta=\pi/3,\ \theta=\pi/4$

90. A

Sol. We have $\sum_{m=1}^{n}\tan^{-1}\left(\dfrac{2m}{m^4+m^2+2}\right)=\sum_{m=1}^{n}\tan^{-1}\left(\dfrac{2m}{1+(m^2+m+1)(m^2-m+1)}\right)$

$=\sum_{m=1}^{n}\tan^{-1}\left(\dfrac{(m^2+m+1)-(m^2-m+1)}{1+(m^2+m+1)-(m^2-m+1)}\right)$

$=\sum_{m=1}^{n}\left[\tan^{-1}(m^2+m+1)-\tan^{-1}(m^2-m+1)\right]$

$=(\tan^{-1}3-\tan^{-1}1)+(\tan^{-1}7-\tan^{-1}3)+(\tan^{-1}13-\tan^{-1}7)+\ldots+$

$\left[\tan^{-1}(n^2+n+1)-\tan^{-1}(n^2-n+1)\right]$

$=\tan^{-1}(n^2+n+1)-\tan^{-1}1=\tan^{-1}\left(\dfrac{n^2+n}{2+n^2+n}\right)$